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IB Mathematics Analysis & Approaches
Higher Level–Prediction Exams
November 2026 - Paper 3



Duration: 1:15 hrs

(Calculator is Allowed)

Max. Marks: 55

Question 1

[Maximum mark: 26]

This question asks you to investigate the n th derivative of xa^x and errors arising from approximations using its Maclaurin series.

Consider the function $f(x) = xa^x$, where $x \in \mathbb{R}$ and $a \in \mathbb{R}$, $a > 0$, $a \neq 1$.

(a) (i) Show that $f'(x) = a^x (1 + x \ln a)$. [1]

(ii) Find $f''(x)$. [2]

(iii) Find $f'''(x)$. [1]

Answer:

$$\begin{aligned} (a) \quad (i) \quad f(x) &= xa^x \\ f'(x) &= 1 \cdot a^x + x \cdot a^x \cdot \ln a \\ \therefore f'(x) &= a^x (1 + x \ln a) \end{aligned} \quad \left\{ \begin{array}{l} (5.6) \\ ((uv)' = u'v + uv') \\ \frac{d}{dx}(a^x) = a^x \cdot \ln a \end{array} \right\}$$

$$(ii) \quad f''(x) = a^x \cdot \ln a (1 + x \ln a) + a^x (\ln a \cdot 1)$$

$$f''(x) = a^x \ln a (2 + x \ln a)$$

$$(iii) \quad f'''(x) = 2(\ln a) a^x + (\ln a)^2 x a^x$$

$$\begin{aligned} \therefore f'''(x) &= 2(\ln a)^2 a^x + (\ln a)^2 [1 \cdot a^x + x a^x \cdot \ln a] \\ &= \underline{2(\ln a)^2 a^x} + \underline{(\ln a)^2 a^x} + x a^x (\ln a)^3 \end{aligned}$$

$$\therefore f'''(x) = (\ln a)^2 a^x (3 + x \ln a)$$

It is conjectured that $f^{(n)}(x) = a^x (\ln a)^{n-1} (n + x \ln a)$, for $n \in \mathbb{Z}^+$.

(b) Prove by mathematical induction that this conjecture is correct, for all $n \in \mathbb{Z}^+$. [8]

(c) Hence find the value of $f^{(n)}(0)$, for $n \in \mathbb{Z}^+$. [2]

(b) Let

$$p(x) : f^{(n)}(x) = a^x (\ln a)^{n-1} (n + x \ln a), \text{ for } n \in \mathbb{Z}^+$$

(i) For $n=1$

$$RHS = a^x (\ln a)^{1-1} (1 + x \ln a)$$

From part (a) it is $f'(x)$

So, $P(1)$ is true

(ii) Now assume that $P(k)$ is true for some, $k \in \mathbb{Z}^+$

$$\therefore f^{(k)}(x) = a^x (\ln a)^{k-1} (k + x \ln a) \rightarrow f^{(k+1)}(x) = a^x (\ln a)^{k-1} (k + x \ln a) + a^x (\ln a)^{k-1} (0 + 1 \cdot \ln a)$$

(iii) Now we need to prove that $P(k+1)$ is true, that is

$$f^{(k+1)}(x) = a^x (\ln a)^k (k + 1 + x \ln a) = a^x (\ln a)^k (k + x \ln a) + a^x (\ln a)^k$$

Differentiating both sides of assumption for $P(k)$, with respect to x , we have

$$f^{(k+1)}(x) = a^x (\ln a)^{k-1} (\ln a) + (a^x \cdot \ln a) (\ln a)^{k-1} (k + x \ln a)$$

$$= a^x (\ln a)^k + a^x \cdot (\ln a)^k (k + x \ln a)$$

$$f^{(k+1)}(x) = a^x (\ln a)^k (k + 1 + x \ln a)$$

Hence $P(k+1)$ is true, whenever $P(k)$ is true

Since $P(1)$ is true and $P(k)$ true $\Rightarrow P(k+1)$ is true, by the method of mathematical induction, $P(n)$ for all $x \in \mathbb{Z}^+$

(c) Substituting $n=1$ in $f^{(n)}(x)$ we get

$$f^{(n)}(0) = a^0 (\ln a)^{n-1} (n + 0)$$

$$f^{(n)}(0) = n (\ln a)^{n-1}$$

(d) Determine the Maclaurin series for xa^x . Give your answer up to and including the term in x^4 .

[6]

By Maclaurin series (5.19), we have

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f^{(2)}(0) + \frac{x^3}{3!}f^{(3)}(0) + \frac{x^4}{4!}f^{(4)}(0) + \dots$$

From the given function $f(0) = 0$

$$(f(x) = x a^x) \\ f(0) = 0 a^0 = 0$$

From part(a), derivatives at $x = 0$ are

$$f'(x) = a^x (1 + x \ln a) \Rightarrow f'(0) = a^0 (1 + 0) = 1$$

(from part (a))

$$f''(x) = a^x \ln a (2 + x \ln a) \Rightarrow f''(0) = a^0 \ln a (2 + 0) = 2 \ln a$$

$$f'''(x) = (\ln a)^2 a^x (3 + x \ln a) \Rightarrow f'''(0) = (\ln a)^2 a^0 (3 + 0) = 3 (\ln a)^2$$

For the fourth order derivative, From part (c), we have

$$f^{(n)}(0) = n (\ln a)^{n-1} \Rightarrow f^{(4)}(0) = 4 (\ln a)^3$$

Substituting these values into the series, we have

$$\begin{aligned} xa^x &= 0 + x(1) + \frac{x^2}{2!}(2 \ln a) + \frac{x^3}{3!} \times 3 (\ln a)^2 + \frac{x^4}{4!} \times 4 (\ln a)^3 + \dots \\ &= x + x^2 \ln a + \frac{x^3}{2} (\ln a)^2 + \frac{x^4}{6} (\ln a)^3 + \dots \end{aligned}$$

Hence Maclaurin series approximation up to and including the term in x^4 , is

$$xa^x \approx x + (\ln a) x^2 + \frac{(\ln a)^2}{2} x^3 + \frac{(\ln a)^3}{6} x^4$$

For the remainder of this question, let $a = 2$, and let $P_n(x)$ denote the Maclaurin series for xa^x up to and including the term in x^n .

Consider the error function $\varepsilon_n(x) = f(x) - P_n(x)$.

(e) (i) Find an expression for $\varepsilon_4(x)$ [2]

(ii) Hence find the range of values of x such that $|\varepsilon_n(x)| < 0.005$ when $n = 4$. [4]

(i) We have

$$\varepsilon_4(x) = f(x) - P_4(x)$$

with $a=2$ from part (d)

$$P_4(x) = x + (\ln 2)x^2 + \frac{(\ln 2)^2}{2}x^3 + \frac{(\ln 2)^3}{6}x^4$$

and

(given)

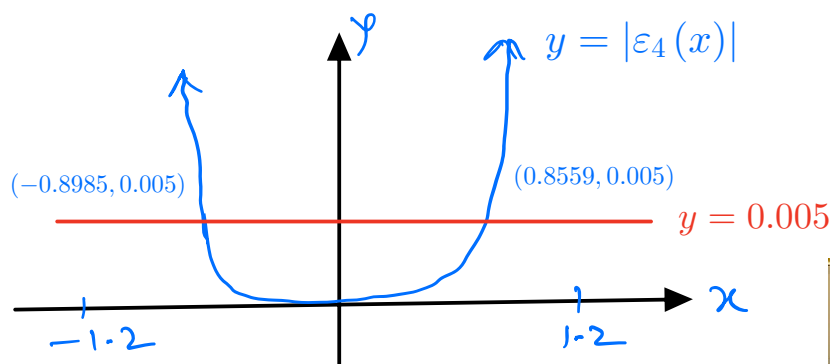
$$f(x) = x(2^x)$$

$$\therefore \varepsilon_4(x) = (x2^x) - x - (\ln 2)x^2 - \frac{(\ln 2)^2}{2}x^3 - \frac{(\ln 2)^3}{6}x^4$$

(ii) Using GDC, the graph of

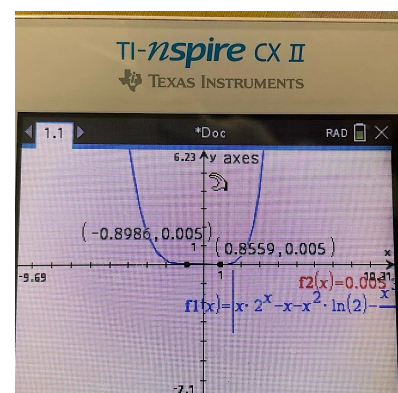
$$y = |\varepsilon_4(x)| \quad \text{and} \quad y = 0.005$$

The points of intersection as shown in graph, are



The interval where $|\varepsilon_n(x)| < 0.005$ is

$$-0.899 < x < 0.856$$



Question 2

[Maximum mark: 29]



This question asks you to investigate the limit of a family of solutions to a first-order linear differential equation, as $x \rightarrow \infty$.

Consider the derivative

$$f'(x) = e^{ax} \cos(bx), \quad a, b \in \mathbb{R}^+.$$

(a) Show that

$$f(x) = \frac{e^{ax}(a \cos(bx) + b \sin(bx))}{a^2 + b^2} + C.$$

where C is an arbitrary constant.

[6]

Answer: From

$$f'(x) = e^{ax} \cos(bx)$$

$$f(x) = \int e^{ax} \cos(bx) dx$$

$$= \int \cos(bx) e^{ax} dx \quad \text{(ILATE)} \quad \left(\int u dv = uv - \int v du \right)$$

$$= \cos(bx) \frac{e^{ax}}{a} - \int (-\sin(bx) \cdot b) \cdot \frac{e^{ax}}{a} dx$$

$$= \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a} \int \sin(bx) e^{ax} dx$$

$$= \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a} \left[\sin(bx) \left(\frac{e^{ax}}{a} \right) - \int (\cos(bx) \cdot b) \left(\frac{e^{ax}}{a} \right) dx \right]$$

$$f(x) = \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a^2} \sin(bx) e^{ax} - \frac{b^2}{a^2} \int \cos(bx) e^{ax} dx$$

$$\underline{f(x)} = \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a^2} \sin(bx) e^{ax} - \frac{b^2}{a^2} \underline{f(x)}$$

$$\therefore \left(1 + \frac{b^2}{a^2} \right) f(x) = \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a^2} \sin(bx) e^{ax} \quad \left(1 + \frac{b^2}{a^2} = \frac{a^2 + b^2}{a^2} \right)$$

$$\left(\frac{a^2 + b^2}{a^2} \right) f(x) = \frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a^2} \sin(bx) e^{ax}$$

Multiplying both sides by $\frac{a^2}{a^2 + b^2}$



$$\therefore f(x) = \frac{a^2}{a^2 + b^2} \left[\frac{1}{a} \cos(bx) e^{ax} + \frac{b}{a^2} \sin(bx) e^{ax} \right]$$

$$f(x) = \frac{a \cos(bx) e^{ax} + b \sin(bx) e^{ax}}{a^2 + b^2}$$

Since f(x) is indefinite integral, therefore

$$f(x) = \frac{e^{ax} [a \cos(bx) + b \sin(bx)]}{a^2 + b^2} + C$$

Now consider the differential equation $\frac{dy}{dx} + y = 2 \cos x$.

(b) (i) Write down the integrating factor for this differential equation. [1]

(ii) Hence show that the general solution to this differential equation is

$$y = \sin x + \cos x + Ce^{-x},$$

where C is an arbitrary constant. [5]

Answer:

$$\frac{dy}{dx} + y = 2 \cos x.$$

Comparing the given differential equation with

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Formula (5.18)

We have

$$P(x) = 1$$

(i) Using the integrating factor

$$I(x) = e^{\int 1 dx} = e^x$$

(ii) Multiplying the D.E. by integrating factor

$$e^x \left(\frac{dy}{dx} + y \right) = e^x (2 \cos x)$$

$$\frac{d}{dx} (e^x y) = 2 \cos x e^x$$

Integrating both sides with respect to 'x', we have

$$\int \frac{d}{dx} (e^x y) dx = 2 \int \cos x e^x dx \quad \left(\int u dv = uv - \int dv \right)$$

$$e^x y = 2 \int \cos x e^x dx \quad \text{--- (1)}$$

Using result from part(a) with $a = 1$ and $b = 1$, we have

$$\int \cos x e^x dx = \frac{e^x (\cos x + \sin x)}{2} + C$$

Substituting in (i), we get

$$e^x y = 2 \left[\frac{e^x (\cos x + \sin x)}{2} \right] + C$$

$$y = (\sin x + \cos x) + ce^{-x} \quad \text{---(ii)}$$

(c) (i) Show that $\sin x + \cos x \equiv \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$. [3]

(ii) Hence, explain why every curve in the family of solutions found in part (b) approaches the curve $y = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$ as $x \rightarrow \infty$, regardless of the value of C . [1]

Answer:

$$(i) \quad \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin \left(x + \frac{\pi}{4} \right) = \sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}$$

$$= \frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x$$

$$= \frac{1}{\sqrt{2}} (\sin x + \cos x)$$

$$\therefore \sqrt{2} \sin \left(x + \frac{\pi}{4} \right) = \sin x + \cos x$$

Part(b)

$$y = (\sin x + \cos x) + ce^{-x}$$

(ii)

General solution can be written as

$$y = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right) + ce^{-x}$$

(from (i))

as $x \rightarrow \infty, Ce^{-x} \rightarrow 0$ Regardless of value of C

Therefore, every curve in the family of solutions approaches

$$y = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right), \text{ as } x \rightarrow \infty$$

It is given that $y = 0$ when $x = 0$.



(d) (i) Find the particular solution satisfying this condition.

[2]

(ii) Explain why this particular solution never actually meets the curve

$$y = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right).$$

[2]

Answer:

(i) Putting $x=0$ and $y=0$ in general solution, we have

$$0 = \sin 0 + \cos 0 + Ce^0$$

$$0 = 1 + c \Rightarrow c = -1$$

Hence, particular solution is

$$y = \sin x + \cos x - ce^{-x}$$

(ii) The vertical distance between this particular solution and the curve

$$y = \sin x + \cos x$$

$$|(\sin x + \cos x - e^{-x}) - (\sin x + \cos x)| = |-e^{-x}|$$

Since $e^{-x} > 0$, for all values of x , we have

$$|-e^{-x}| = e^{-x}$$

Hence, the vertical distance between the two curves is never zero, so particular solution never meets this curve.

The differential equation is now generalized to $\frac{dy}{dx} + ky = A \cos x$, where k and A are positive constants.

(e) (i) Show that the general solution to this differential equation can be written as

$$y = \frac{A(\sin x + k \cos x)}{k^2 + 1} + Ce^{-kx}.$$

[3]

Answer:

(i) Integrating factor is $I(x) = e^{\int k dx} = e^{kx}$

$$e^{kx} \left[\frac{dy}{dx} + ky \right] = e^{kx} [A \cos x]$$

$$\frac{d}{dx} (e^{kx} y) = A \cos x e^{kx}$$

Integrating both sides with respect to x , we get

$$\therefore e^{kx} y = \int A \cos x e^{kx} dx$$

Using result from part (a) with $a = k$ and $b = 1$, we have

$$e^{kx} y = A \left[\frac{e^{kx} (k \cos x + \sin x)}{k^2 + 1} + C \right]$$

Consider the derivative

(a) Show that

$$f'(x) = e^{ax} \cos(bx), \quad a, b \in \mathbb{R}^+.$$

$$f(x) = \frac{e^{ax} (a \cos(bx) + b \sin(bx))}{a^2 + b^2} + C.$$

$$y = A \frac{(\sin x + k \cos x)}{k^2 + 1} + Ce^{-kx}$$

- (ii) Using the method from part (c), express $\sin x + k \cos x$ in the form $R \sin(x + \alpha)$, $R > 0$, and hence determine the amplitude of the function approached by this family of curves as $x \rightarrow \infty$.

[6]

Answer: In (c) part (i)

$$\sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$$

where amplitude is $\sqrt{2}$

Now using compound angle identity

$$\begin{aligned} R \sin(x + \alpha) &= R \sin x \cos \alpha + R \cos x \sin \alpha \\ &= (R \cos \alpha) \sin x + (R \sin \alpha) \cos x \end{aligned}$$

Taking $R \cos \alpha = 1$ and $R \sin \alpha = k$

$$R^2 \cos^2 \alpha = 1, R^2 \sin^2 \alpha = k^2$$

$$\therefore R^2 (\cos^2 \alpha + \sin^2 \alpha) = 1 + k^2$$

$$R^2 = 1 + k^2$$

$$\therefore R = \sqrt{1 + k^2} \quad (R > 0)$$

$$\therefore \sin x + k \cos x = \sqrt{1 + k^2} \sin(x + \alpha)$$

$\therefore$ as $x \rightarrow \infty$, The family of curves approaches

$$\begin{aligned} \frac{A(\sin x + k \cos x)}{k^2 + 1} &= \frac{A\sqrt{k^2 + 1} \sin(x + \alpha)}{k^2 + 1} \\ &= \frac{A \sin(x + \alpha)}{\sqrt{k^2 + 1}} = \frac{A}{\sqrt{k^2 + 1}} \sin(x + \alpha) \end{aligned}$$

and $ce^{-kx} \rightarrow 0$, as $x \rightarrow \infty$

Hence amplitude of function approaches

$$\frac{A}{\sqrt{k^2 + 1}}$$

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$\int_a^b f(x) dx$
 $\frac{d}{dx} [f(x)]$
 $\sum_{k=0}^n \binom{n}{k}$
 $P(A|B) = \frac{P(A \cap B)}{P(B)}$
 $\vec{a} \cdot \vec{b}$



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